A rational counterexample to pentagon barycenter coincidence

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Abstract

We give a strictly convex pentagon with rational vertices on the unit circle whose hyperbolic barycenter and Euclidean least-squares point coincide away from the origin, although its five consecutive cross-ratios are not equal. This disproves Conjecture 36 of Arnold and Arreche.

Task 1: the original problem

We use the notation of the actual source [1]: $S_{\mathbf{P}}$ is the hyperbolic barycenter and $L_{\mathbf{P}}$ minimizes the sum of squared Euclidean distances to the side lines. Both are expressed in the Beltrami–Cayley–Klein unit-disc coordinates. Conjecture 7.8 of the preprint, retained as Conjecture 36 of the journal final, asserts that $S_{\mathbf{P}} = L_{\mathbf{P}}$ for a pentagon if and only if its five consecutive cross-ratios are equal or $S_{\mathbf{P}} = (0, 0)$.

Theorem 1. *Let the vertices of* $\mathbf{P}$*, in cyclic order, be*

$$P_1 = (1, 0), \quad P_2 = \left(\frac{7}{25}, \frac{24}{25}\right), \quad P_3 = \left(-\frac{4}{5}, \frac{3}{5}\right), \quad P_4 = \left(-\frac{4}{5}, -\frac{3}{5}\right), \quad P_5 = \left(\frac{7}{25}, -\frac{24}{25}\right).$$

Then $\mathbf{P}$ *is strictly convex and*

$$S_{\mathbf{P}} = L_{\mathbf{P}} = \left(-\frac{1}{62}, 0\right),$$

but its five consecutive cross-ratios are not all equal.

The polygon and the source barycenter formula

The source's stereographic parameters give

$$P(p) = \left(\frac{1 - p^2}{1 + p^2}, \frac{2p}{1 + p^2}\right), \qquad (p_1, p_2, p_3, p_4, p_5) = \left(0, \frac{3}{4}, 3, -3, -\frac{3}{4}\right).$$

These are five distinct points on the unit circle in counterclockwise order, so their polygon is strictly convex. With indices modulo five, the source defines

$$I = \sum_i \frac{1}{p_i - p_{i+1}}, \qquad J = \frac{1}{2} \sum_i \frac{p_i + p_{i+1}}{p_i - p_{i+1}}, \qquad K = \sum_i \frac{p_i p_{i+1}}{p_i - p_{i+1}}.$$

Its equation (3.3) and Theorem 5 in the journal final give

$$S_{\mathbf{P}} = \left(\frac{I - K}{I + K}, \frac{2J}{I + K}\right).$$

Direct substitution gives $I = -61/18$, $J = 0$, $K = -7/2$, and hence $I + K = -62/9$ and $I - K = 1/9$. Therefore $S_{\mathbf{P}} = (-1/62, 0)$.

The Euclidean least-squares point

The five side lines, in order, are

$$4x + 3y = 4, \quad -5x + 15y = 13, \quad 5x = -4, \quad -5x - 15y = 13, \quad 4x - 3y = 4.$$

Thus their squared-distance sum is

$$\begin{aligned} F(x,y) &= \frac{(4x+3y-4)^2 + (4x-3y-4)^2}{25} + \frac{(5x+4)^2}{25} \\ &\quad + \frac{(-5x+15y-13)^2 + (-5x-15y-13)^2}{250} \\ &= \frac{62}{25}x^2 + \frac{63}{25}y^2 + \frac{2}{25}x + \frac{409}{125}. \end{aligned}$$

Its Hessian is $\mathrm{diag}(124/25, 126/25)$, which is positive definite. Its unique global minimum is consequently at $(-1/62, 0)$. This proves $L_{\mathbf{P}} = S_{\mathbf{P}} \neq (0,0)$.

The unequal cross-ratios

To test equality, fix the ordered convention

$$R(a,b,c,d) = \frac{(a-c)(b-d)}{(a-d)(b-c)}, \qquad R_i = R(p_i, p_{i+1}, p_{i+2}, p_{i+3}).$$

The five values are

$$(R_1, R_2, R_3, R_4, R_5) = \left(\frac{5}{3}, \frac{25}{16}, \frac{5}{3}, \frac{8}{5}, \frac{8}{5}\right).$$

In particular $R_1 \neq R_2$. Changing a fixed ordered cross-ratio convention applies an injective fractional-linear function (possibly with a change of sign) to each value, so it does not affect whether the five values are equal. This disproves the necessary direction of the conjecture. □

An independent hyperbolic check

For a chord line $ax + by = c$ in the unit-disc model, the hyperbolic distance d_H satisfies

$$\sinh d_H((x,y), \{ax + by = c\}) = \frac{|ax + by - c|}{\sqrt{a^2 + b^2 - c^2}\sqrt{1 - x^2 - y^2}}.$$

This follows directly from the polar and hyperboloid formulas (2.5)–(2.7) of [1]. Choose the inward linear forms

$$q_1 = 4 - 4x - 3y, \quad q_2 = 13 + 5x - 15y, \quad q_3 = 4 + 5x, \quad q_4 = 13 + 5x + 15y, \quad q_5 = 4 - 4x + 3y.$$

The corresponding normalizing factors are $(3, 9, 3, 9, 3)$, so throughout the open disc the hyperbolic objective $G = \sum_i \sinh d_{H,i}$ obeys

$$G(x,y) \geq \frac{62 + x}{9\sqrt{1 - x^2 - y^2}} \geq \frac{\sqrt{3843}}{9}.$$

The second inequality, including its equality case, follows from

$$(62 + x)^2 - 3843(1 - x^2 - y^2) = (62x + 1)^2 + 3843y^2$$

and $62 + x > 0$. At $(-1/62, 0)$ all five q_i are positive and both inequalities are equalities. This independently certifies the same unique hyperbolic minimum, with no appeal to a numerical optimizer.

Scope and verification

The example satisfies convexity, cyclic order, distinctness, and the unit circumcircle normalization. It refutes the source statement itself; the abbreviated MathDB names for the two centers are not used as a loophole. The companion verifier uses exact arithmetic for the geometry, both center calculations, the hyperbolic minimum certificate, and the cross-ratios. Review is by the authoring agent only, without independent peer review or a proof-assistant certificate.

Proposed follow-up Tasks 2–5

Task 2: Classify coincidence in the reflection-symmetric family $p = (0, a, b, -b, -a)$ with $0 < a < b$.

Task 3: Determine the connected components of all strictly convex pentagons with $S_{\mathbf{P}} = L_{\mathbf{P}} \neq 0$, modulo rotations of the unit disc.

Task 4: Characterize which strictly convex ideal pentagons with $S_{\mathbf{P}} = 0$ also have $L_{\mathbf{P}} = 0$.

Task 5: For every $n \geq 6$, determine whether a strictly convex rational unit-circle n-gon exists with coincident nonzero centers and unequal consecutive cross-ratios.

References

[1] M. Arnold and C. E. Arreche, *Symmedians as Hyperbolic Barycenters*, Comptes Rendus Mathématique **362** (2024), 1743–1762. Journal version (DOI). Preprint: arXiv:2311.14194v1.